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Summer Term 2026

Pattern Matching and Machine Learning for Audio Signal Processing
Exercise sheet 10

To be uploaded in eCampus till: 19-07-2026 22:00 (strict deadline)

This is the last exercise sheet of this term. The exercises have a level of difficulty similar to the tasks you will have to solve in the final exam. With these exercises you can gain bonus points.

Exercise 10.1 [3 points]

Consider the complex number $z = \left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)^3$. Choose the correct answer(s) between the ones given below. Prove your answers.

- ☐ z can also be written as $z = e^{\frac{\pi}{4}i}$.
- ☐ z can also be written as $z = e^{\frac{3}{4}\pi i}$.
- ☐ z has a length of 2.

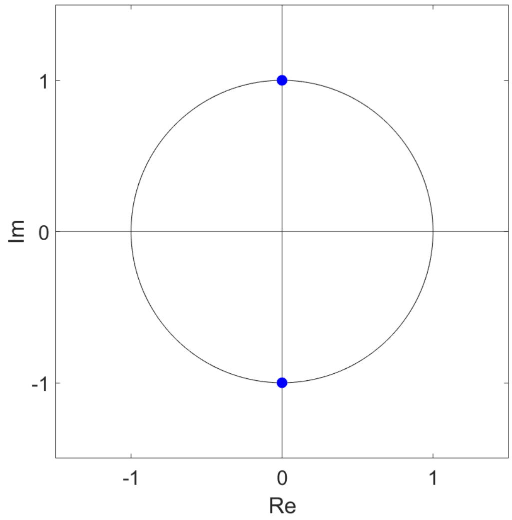
Exercise 10.2 [2 points]

Let $\mathbf{x} = (3, 0, 2, 1)^T$. Calculate the spectrum $\mathbf{X}$.

Exercise 10.3

[2 points]

Indicate which of the following statements about the figure illustrated below is correct.



The figure shows the...

- [ ] ...2nd roots of unity.
- [ ] ...primitive 2nd roots of unity.
- [ ] ...primitive 4th roots of unity.
- [ ] ...primitive 8th roots of unity.

Exercise 10.4

[2 points]

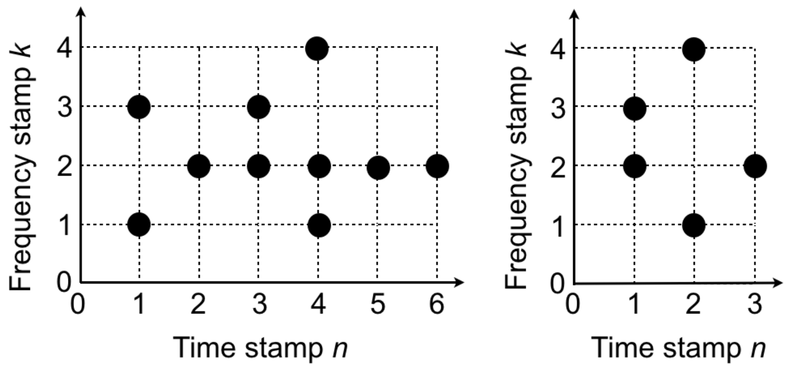
Choose the correct answers.

- [ ] The cyclic autocorrelation of a signal of length N has length $N+1$.
- [ ] One can use the spectrum of a signal to calculate its autocorrelation.
- [ ] When calculating the STFT of a signal, a short window means good frequency resolution.
- [ ] Percussions can be seen in the spectrogram (with time on the x-axis and frequency on the y-axis) of a music signal as horizontal lines.

Exercise 10.5

[$2 + 1 = 3$ points]

Consider the constellation maps $C(\mathcal{D})$ (left) and $C(\mathcal{Q})$ (right) as specified by the following figure.



(a) Compute the corresponding inverted lists and the indicator functions.

(b) Compute the resulting matching function.

Exercise 10.6

[$1 + 2 = 3$ points]

(a) Write down the definition of the log frequency spectrogram.

(b) Explain the steps you need to compute the chroma energy normalized statistics (CENS_d^l).

Exercise 10.7

[2 points]

Explain a simple template-based chord recognition strategy based on binary chord templates.